Temporal Processing in L1 and L2 Chinese: A Visual World Eye-Tracking Study

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Previous research has shown that native (L1) speakers of Spanish[1], Russian[2], English[3] and (Mandarin) Chinese[4] use aspectual morphemes as cues to construct event representations during real-time sentence comprehension. However, it remains unknown whether second language (L2) learners can integrate aspectual morphemes in a similar manner. Furthermore, no previous studies of either L1 or L2 processing have examined the role of temporal adverbs as cues during online processing. The current study fills these gaps by investigating how L1 and L2 Chinese speakers integrate temporal information encoded in the postverbal aspectual morphemes (perfective –*le*, imperfective –*zhe*) and preverbal temporal adverbs (e.g., *yijing* 'already', *kuaiyao* 'about to')[5] to identify event types that the sentence describes during listening and how individual differences (in working memory capacity and L2 proficiency) modulate L2 processing. Following the Lexical Bottleneck Hypothesis[6], which posits that slower lexical access hinders or delays structure building in L2, we hypothesized that both groups would use temporal cues online to facilitate event recognition, but L2 speakers might show lower accuracy or delayed integration of these cues.

Methods We recruited 28 L1 Chinese speakers, and 46 L2 Chinese speakers from diverse L1 backgrounds at intermediate to advanced Chinese proficiency. Participants took part in a webcam visual world eye-tracking study on Gorilla, in which they listened to 20 sentences containing different temporal cues and were asked to select the image that best matched what they heard (see Table 1). They heard 9 sentences containing either the perfective -*le*(PFV) or the imperfective -*zhe* (IMPV) while viewing images of completed and ongoing events, and 11 sentences containing *yijing* 'already' (PAST) or *kuaiyao* 'about to' (FUTURE) while viewing images representing past and future events. Participants' picture selection choices, reaction times and eye movements were recorded. L2 speakers also completed a Chinese proficiency test, a backward digit span task, a language background questionnaire, and a sentence interpretation task to measure their working memory capacity, proficiency, and knowledge of the temporal morphemes used in the experimental stimuli, respectively.

Results & Discussion L2 speakers showed near-ceiling accuracy in the sentence interpretation task (Mean: 19.24 out of 20, SD: 1.57), indicating successful acquisition of Chinese temporality marking. For picture selection, L2 speakers were significantly less accurate (ps < 0.05) (Figure 1) and also took significantly longer to select the target picture than L1 speakers (ps < 0.05) (Figure 2) in the FUTURE and IMPV conditions. The observed L1–L2 difference in IMPV aligns with Aspect Hypothesis [7] as imperfective aspect combined with accomplishment (lexical aspect) poses greater difficulty for L2 speakers. Moreover, FUTURE and IMPV are less frequent than PAST and PFV based on Chinese CCL corpus [8] and previous corpus study [9], so insufficient L2 input limits their automatic processing skills. Individual differences analysis found that L2 proficiency, but not working memory, predicted L2's reaction time, with higher-proficiency learners showing faster temporal information integration. In the eye movement pattern (Figure 3), we used cluster-based permutation analysis to identify significant cluster where participants' fixation on target picture was above chance. We found that both L1 and L2 speakers can integrate aspectual morphemes and temporal adverbs in real time and showed significantly more looks toward the target images. This finding indicates that they can incrementally integrate these temporal cues during listening to construct an event representation before the sentence ends. We also conducted bootstrapped analysis to directly compare the onset of divergence between the L1 and L2 speakers in each condition. In the PAST condition only, L1 speakers showed significantly earlier divergence onset than L2 speakers (Δ = 835 ms, 95% CI = [600, 1100]). No significant group differences were found in the other conditions. Together with the significant effect of Chinese proficiency on picture selection RTs, these eye fixation patterns provide partial support for the Lexical Bottleneck Hypothesis.

Table 1 Sample material and visual display

Aspectual morphemes	Temporal adverbs

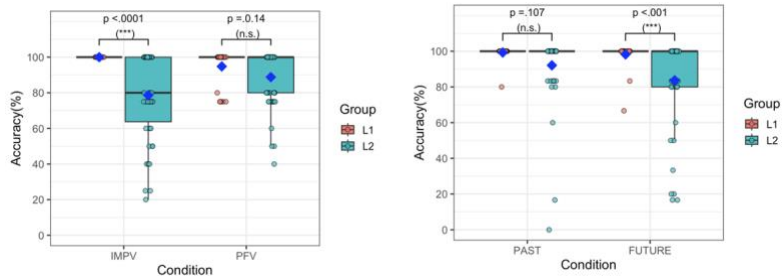

Figure 1 Mean accuracy (%) across conditions for L1 and L2 speakers. (Diamonds = group means; Boxes = IQR; Horizontal lines = medians)

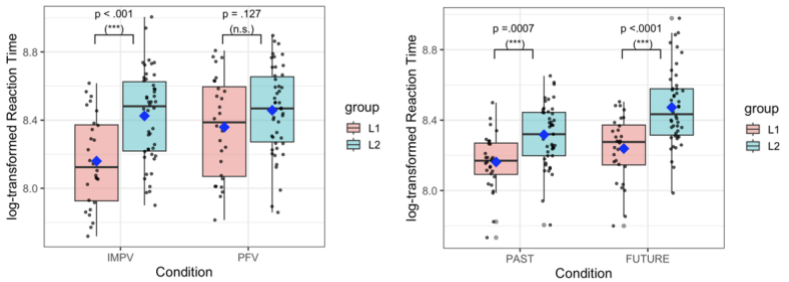

Figure 2 Mean reaction time across conditions for L1 and L2 speakers. (Diamonds = group means; Boxes = IQR; Horizontal lines = median)

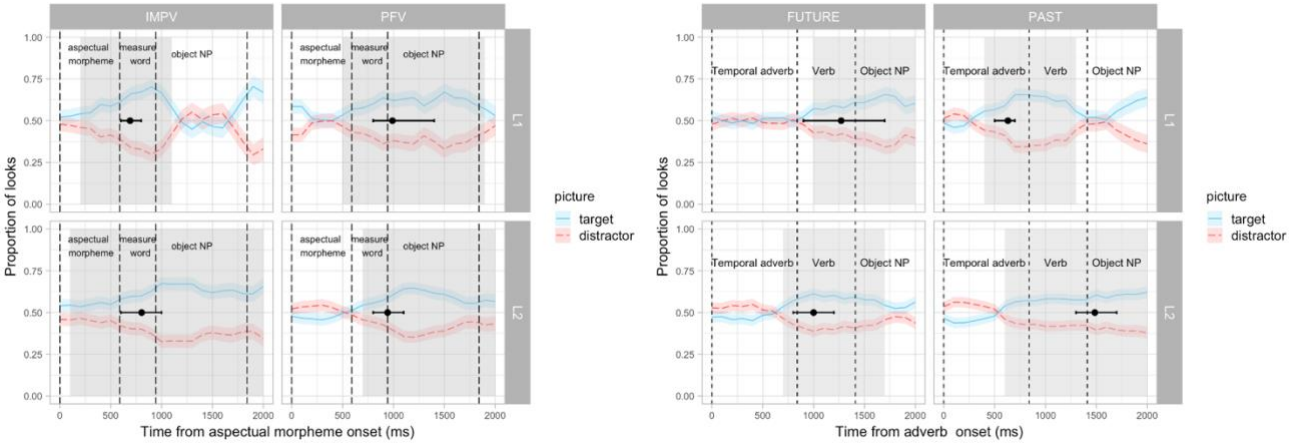

Figure 3 Proportions of looks to target (blue solid) and distractor pictures (red dashed) by conditions and groups, time-locked to the onset the aspectual morphemes or adverbs.
The vertical dashed lines represent the onset of different sentence elements.
The grey shaded areas represent significant clusters identified via a cluster-based permutation analysis.
The black dots and horizontal bars mark the estimated divergence point and its 95% bootstrap CI.

References

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